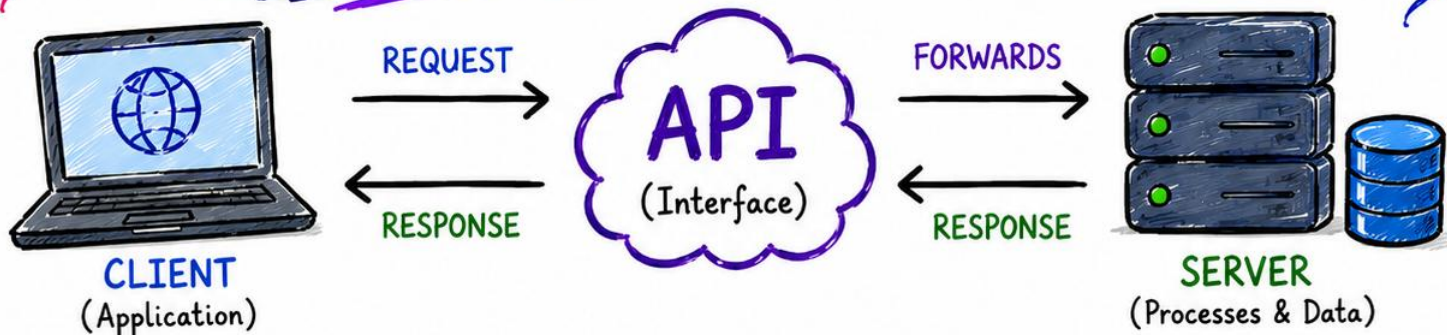


API

IN DEPTH



BUILD



INTEGRATE



INNOVATE



SECURE

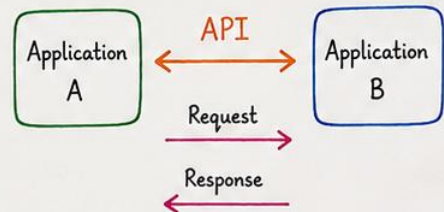


SCALE

API (Application Programming Interface)

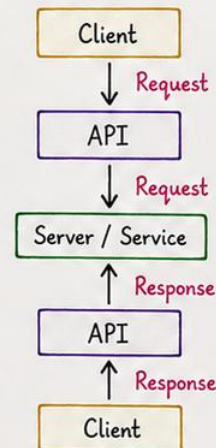
✓ 1. What is API?

API (Application Programming Interface) is a set of rules and protocols that allows two software applications to **communicate with each other**. It acts as a **bridge** between different systems so that they can share data and functionality.



✓ 2. How does API work?

- The client (application) sends a **request** to the API.
- The API takes the request to the server.
- The server **processes** the request.
- The server sends back a **response** (usually data) to the API.
- The API sends the **response** back to the client.
- The client uses that data or functionality in its application.



✓ 3. Why is API used?

- Allows different software systems to **communicate**.
- Promotes **reusability** – don't build everything from scratch.
- Saves **time** and **effort**.
- Provides **security** by controlled access.
- Enables **integration** of third-party services (e.g., payment, maps, login, weather, etc.).

Real-life Example

When we log in using Google in an app, the app uses **Google's** API to verify our identity.

✓ 4. Key Points to Remember

- ★ API is not the software, it is a **connector**.
- ★ It defines how **requests** are made and how **responses** are received.
- ★ It helps apps work together **smoothly**.

In Simple Words

API is like a waiter in a restaurant. You (the client) place an order (**request**), the waiter takes it to the kitchen (server) and brings back the food (**response**).



API (Application Programming Interface)

✓ 5. Types of API

① Public API (Open API)

- Available to everyone over the internet.
- Example: Google Maps API, X API

② Private API

- Used within an organization or company.
- Not exposed to the public.
- Example: Internal HR system API

③ Partner API

- Shared with specific partners or third parties.
- Requires authentication and access approval.
- Example: Payment gateway APIs for merchants

④ Composite API

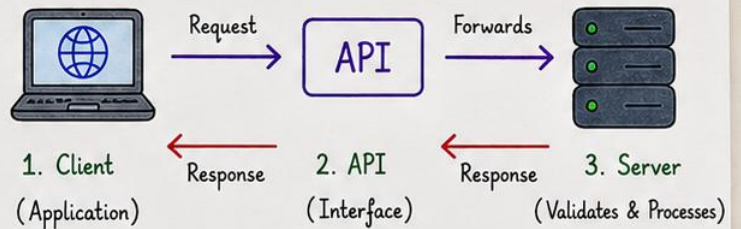
- Combines multiple APIs to perform a larger task.
- Example: Travel app using Flight API + Hotel API + Payment API

★ Common API Protocols ★

- **REST** - Most popular, uses HTTP methods.
- **SOAP** - Uses XML, more rigid and heavy.
- **GraphQL** - Allows clients to request only the data they need.



✓ 6. How does API Work? (Flow)



The client sends a **request** → API **forwards** it to the server → Server **validates & processes** it → API sends back **response** to the client.

✓ 7. Common HTTP Methods Used in APIs

Method	Purpose	Example
GET	Retrieve data	GET /users
POST	Create new data	POST /users
PUT	Update existing data	PUT /users/1
DELETE	Delete data	DELETE /users/1
PATCH	Partial update	PATCH /users/1

✓ 8. Benefits of Using APIs

- ★ Promotes **reusability** and saves development time.
- ★ Enables **communication** between different systems.
- ★ Encourages **scalability** and **flexibility**.
- ★ Allows **integration** with third-party services.
- ★ Helps businesses **innovate** faster.

★ APIs make applications **smarter, faster and connected!**

API (Application Programming Interface)

✓ 9. Real-life Examples of APIs

- ★ Google Maps API - to show maps, locations, directions.
- ★ Payment Gateway API - to process online payments.
- ★ Weather API - to get weather information.
- ★ Social Media APIs - to login, share posts, fetch data.
- ★ Email API - to send emails from applications.

APIs are everywhere!

From mobile apps, websites to systems, APIs power the communication behind the scenes.

✓ 10. Advantages of APIs

- ✓ Reusability of code and functionality.
- ✓ Saves time and development cost.
- ✓ Enables integration with other systems.
- ✓ Improves scalability and performance.
- ✓ Supports innovation and new features.



✓ 11. Challenges of APIs

- ★ Security vulnerabilities (if not implemented properly).
- ★ Dependence on third-party services.
- ★ Versioning and compatibility issues.
- ★ Rate limits and usage restrictions.
- ★ Proper documentation is required.



✓ 12. API Lifecycle



✓ 13. Key Takeaway

APIs are the building blocks of modern applications. They **connect**, **communicate** and **collaborate** to deliver powerful digital experiences.

- ≥ Think API,
- Think Integration,
- ✓ Think Possibilities! ★